

Literature Review: Policy Search Under Combined Observation Noise and Reward Sparsity

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Abstract Real world deployments of policy search algorithms face two difficulties that current research seldom combines: corrupted observations and infrequent rewards. This literature review consolidates and analyses the evidence base for a study comparing Proximal Policy Optimization (PPO), a gradient-based method, against Covariance Matrix Adaptation Evolution Strategy (CMA-ES), a gradient-free population-based method, under both stressors simultaneously on continuous-control tasks. I overview these algorithm’s foundations by their original design papers, survey the disjoint bodies of work that address observation noise and reward sparsity, and show that these two strands have never been formally combined in comparison to two algorithms in my specific context. The literature on both sparsity and noise in our context remains moderately sparse, particularly on the sparsity side, lacking definitive benchmarks.

1 Introduction and Scope

Deploying policy search algorithms in real world applications such as robotics and autonomous vehicles means handling conditions that idealised benchmarks abstract away. Two of the most common of these conditions are corrupted sensor observations and infrequent reward signals. The evolutionary computation community has studied how evolution strategies behave under noisy fitness evaluations for at least two decades, and the reinforcement-learning (RL) community has built a substantial programme on escaping sparse-reward regimes. These two areas, have however, developed almost independently.

This review positions a factorial study comparing PPO [35], a gradient-based policy-gradient method, against CMA-ES [13, 14, 12], a gradient-free population-based method, under combined additive gaussian observation noise and reward sparsity on continuous-control tasks. The two algorithms are deliberately chosen to act as canonical representatives of the two paradigms with fundamental structural contrast ;per-timestep gradient updates against rank-based episode-level selection.

2 Search Scope and Methodology

2.1 Search Strategy

The literature was gathered through a structured search of Google Scholar covering databases such as IEEE Xplore, the ACM Digital Library, arXiv and Scopus, covering 1996 (CMA-ES’s introduction) through early 2026. Five core queries were used, each targeting a specific combination of the core concepts:

- Q1: (PPO OR “Proximal Policy Optimization”) AND (“CMA-ES” OR “Covariance Matrix Adaptation”) AND (“continuous control” OR robotics OR Mujoco OR PyBullet) AND (“sparse reward” OR “delayed reward” OR “binary reward”) AND (“observation noise” OR “sensor noise” OR “Gaussian noise”)
- Q2: (“evolution strategies” OR “CMA-ES” OR “natural ES”) AND (“policy gradient” OR PPO OR TRPO OR RL) AND (“sparse reward” OR “delayed reward” OR “binary reward”) AND (“observation noise” OR “sensor noise” OR “Gaussian noise”)
- Q3: (PPO OR “Proximal Policy Optimization”) AND (“sparse reward” OR “sparse rewards” OR “delayed reward” OR “binary reward”)
- Q4: (PPO OR “Proximal Policy Optimization”) AND (“observation noise” OR “sensor noise” OR “measurement noise”) AND (“additive Gaussian” OR Gaussian OR stochastic)
- Q5: (“CMA-ES” OR “Covariance Matrix Adaptation”) AND (“sparse reward” OR “sparse

rewards” OR “delayed reward”)

Q1, targeting the combined PPO–CMA-ES comparison under both stressors, returned zero relevant results – the direct evidence for the gap claim. Q2 returned a small set of hybrid papers (Section 7) but none crossing both stressors. Q3 dominated by the Ecoffet comparison [6]; Q4 returned relevant papers on PPO and noise by Rahn et al. [32] and Li [23]; Q5 by Ecoffet et al. and preference-based CMA-EPPS [44]. The original and early algorithmic papers gathered with separate queries

2.2 Inclusion Criteria

Papers were included if they (a) studied PPO, CMA-ES, or a hybrid of the two, provided foundational algorithm descriptions, or contributed methodological recommendations; (b) studied observation noise or reward sparsity; and (c) addressed continuous control or (for foundational CMA-ES work) optimisation benchmarks transferable to continuous control.

3 Algorithmic Foundations

CMA-ES and PPO occupy opposite ends of three load-bearing axes that recur throughout the review: gradient-based versus gradient-free; per-timestep versus episodic; and value-function-dependent versus rank-based. Establishing these distinctions precisely from the design papers makes the later mechanisms – and the gaps in their empirical validation – follow directly.

3.1 CMA-ES: Derandomization, Cumulation, and Covariance Adaptation

CMA-ES’s central design idea is derandomization: strategy parameters are updated deterministically from selection information rather than by direct random mutation. Hansen and Ostermeier introduce this via the evolution path [13], motivated by the principle that, in their words, “reasonable adaptation has to reduce the difference between the distributions of the actual evolution path and an evolution path under random selection, with respect to the parameters adapted.” A search should not be viewed as a sequence of independent steps. Instead, it is the path a population takes over a number of generations. This path perspective is

what makes cumulation—and therefore covariance adaptation—possible.

The complete derandomised formulation [14] puts the matter directly: “the underlying objective of mutative strategy parameter control is roughly to favor previously selected mutation steps in the future. If this objective is pursued rigorously, a completely derandomized self-adaptation scheme results.” Cumulation is what makes this stable: aggregated search directions suppress the per-generation stochasticity that would otherwise leak into the covariance estimate, and the paper reports a speed-up factor ranging from three to ten on moderately mis-scaled functions and up to several orders of magnitude on badly scaled nonseparable functions [14]. The default population-size formula $\lambda = 4 + \lfloor 3 \ln n \rfloor$, the weighted recombination weights $w_i = \ln(\mu + \frac{1}{2}) - \ln i$, and the cumulation constants are all set in this paper.

Hansen, Müller and Koumoutsakos [12] then introduce the rank- μ update, which aggregates information from multiple selected candidates per generation rather than relying on the evolution path alone, and reduces the algorithm’s generation-count scaling from roughly $O(n^2)$ in λ to $O(n)$ when $\lambda \propto n$. The motivation they give for treating large populations as a first-class object is explicit and load-bearing for everything that follows about noise. Large populations, they argue, are desired “(1) to reduce the effect of noise; (2) to improve global search properties; and (3) to implement the algorithm on (highly) parallel machines.” Population size is therefore motivated as a noise-tolerance mechanism, not merely observed to correlate with one. All three foundational CMA-ES papers test exclusively deterministic, dense-fitness synthetic functions (Sphere, Ellipsoid, Cigar, Rosenbrock, Rastrigin and others, with n up to 128).

Noise tolerance: design, not measurement.

A caveat that the rest of this review builds on: none of the three foundational CMA-ES papers tests noise empirically. The phrasing “presumably they change in favor of larger values of λ for noisy functions” [12] is a hypothesis, not a measurement. Nor is an RL algorithm of any form covered. The case for CMA-ES’s noise tolerance therefore rests on mechanism design, not on a noise benchmark—a gap the rest of this review traces through every subsequent source.

Beyer and Sendhoff’s MA-ES paper [4] reduces CMA-ES by eliminating the explicit covariance matrix altogether, retaining only its matrix square root M , and shows that the simplified algorithm ”performs nearly as well as the original CMA-ES” on the noiseless BBOB testbed. They confirm the noise-as-motivation framing—”the standard population sizing according to (21) is not well suited for multimodal and noisy optimization” [4]—while again only ever benchmarking on the noiseless BBOB suite.

3.2 PPO: Clipped Policy Gradient Optimisation

PPO [35] optimises a clipped surrogate objective,

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t) \right],$$

where $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$. The clipping mechanism is the algorithm’s central stability device: an ablation in the original paper shows removing clipping and any KL penalty drops the average normalised score across seven MuJoCo environments to -0.39 —worse than a random policy—while $\varepsilon = 0.2$ achieves the best normalised score of 0.82 against 0.76 at $\varepsilon = 0.1$ and 0.70 at $\varepsilon = 0.3$ [35]. Critically, as the paper states plainly, clipping is a step-size constraint; it controls how far an update can push $r_t(\theta)$ but says nothing about whether the gradient direction itself is meaningful. If the gradient direction is corrupted, clipping a biased step produces only a smaller biased step.

PPO combines this with three further mechanisms. Generalized Advantage Estimation (GAE) [34] computes advantages via a λ -interpolation between Monte Carlo returns and one-step TD residuals, $\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V$, where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$, with $\lambda \in [0.9, 0.99]$ recommended in practice. GAE exists for variance reduction: ”GAE($\gamma, 1$) is γ -just regardless of the accuracy of V , but it has high variance due to the sum of terms. GAE($\gamma, 0$) is γ -just for $V = V^{\pi, \gamma}$ and otherwise induces bias, but it typically has much lower variance” [34]. A learned value-function baseline centres the advantages around zero, and multi-epoch minibatch reuse extracts additional sample efficiency by performing K passes of SGD over each trajectory batch. The standard MuJoCo hyperparameters are $T = 2048$,

Adam learning rate 3×10^{-4} , $K = 10$ epochs, mini-batch 64, $\gamma = 0.99$ [35].

The GAE paper additionally flags that long-range dependencies between actions and rewards are a noise source the advantage estimator must work to suppress: it recommends introducing a steeper discount $\gamma\lambda$ to ”cut off the noise arising from long delays” [34]—an explicit acknowledgement that delayed or sparse reward behaves like noise in the gradient estimator. None of PPO’s original experiments, however, tests observation noise, and reward sparsity is mentioned only informally: PPO scores 0.0 on Montezuma’s Revenge [35], the canonical sparse-reward Atari game, but the paper does not frame that as a sparsity result. The PPO paper establishes the algorithm’s mechanics, not its behaviour under either of the stressors this study targets.

3.3 The Algorithmic Divide, Restated

The structural contrast that every subsequent observation in this review follows from is now precise. PPO updates parameters per-timestep via stochastic gradient ascent on a per-timestep gradient estimate; its variance reduction (clipping, GAE, value baseline, multi-epoch reuse) operates on per-timestep quantities. CMA-ES updates a search distribution once per generation from the ranked fitnesses of a population of episodes; its variance reduction (rank-based selection, population averaging, cumulation, covariance adaptation) operates on a single scalar per episode. Any environmental stressor that corrupts per-timestep signals will therefore enter the two algorithms through entirely different pathways—which is exactly why their interaction behaviour is not predictable from either strand of literature alone.

4 Methodological Landscape

Beyond the heavily reviewed primary sources, the experimental design draws on a further body of source material for narrowly scoped methodological choices—hyperparameters, metrics, benchmarks, seed counts and statistical recommendations—rather than independent empirical findings. These are reported here at the level of detail available: what value or definition each source contributes, not a full account of its results.

Hyperparameters. PPO’s learning rate (3×10^{-4}), clipping ($\epsilon = 0.2$) and entropy coefficient (0.01) are standard MuJoCo defaults originally from [35]; the GAE parameter $\lambda = 0.95$ is recommended in [34]. The recent ERL survey by Zhu et al. [45] corroborates these as the de facto MuJoCo defaults across a wide range of comparisons and additionally codifies $\gamma = 0.99$, replay buffers of 10^6 , and actor–critic learning rates that bracket the 3×10^{-4} figure for TD3-style gradient methods. For CMA-ES, the default population-size formula $\lambda = 4 + \lfloor 3 \ln n \rfloor$ and remaining defaults come straight from [14], with [4] re-deriving the cumulation constants in a cleaner form.

Environments. HalfCheetah, Hopper and Walker2d are the three most common MuJoCo benchmarks across this entire literature [45, 32, 6]. Nisioti et al. [27] document the same benchmark dominance and extend it to Brax-based morphological transfer; the TinyverseGP framework [20] formalises the cross-domain vision that puts MuJoCo on the same footing as symbolic regression and logic synthesis for community-driven benchmarking. The Ibehej et al. [19] investigation of inherent structural bias in BBOB and CEC provides an important caveat to interpretation of any benchmark-suite result: BBOB shows a measurable centre bias at high dimension ($\alpha = P(Q_D < 0) = 0.081$ at $D = 40$), substantially more than CEC ($\alpha = 0.25$ – 0.28). Algorithms that approximate central evaluation—as many ES variants implicitly do because (μ, λ) recombination pulls the mean toward the centre of the search space—can appear artificially superior on BBOB at high dimension and low function-evaluation settings. Our study sidesteps this concern by using MuJoCo’s per-environment observation ranges to set noise levels (rather than BBOB), but the finding motivates reporting the precise environment configuration alongside algorithm rankings, rather than collapsing them across multiple benchmark families.

Metrics. Final return—the mean over the last 10% of episodes—is the near-universal primary metric across both RL [35, 45] and ERL [6]. For robustness to observation noise specifically, Li [23] defines the robustness ratio $\varphi_{12} = \mu_R(o_1)/\mu_R(o_2)$ as the ratio of mean returns obtained by a policy trained under one observation-noise distribution and tested under another—the only systematic observation-noise transfer metric in the corpus—and

motivates a rollout-reward standard deviation σ_R as a direct stability metric. The recent continuous-control work of Rahn et al. [32] contributes the concept of a noisy neighborhood—the distribution of returns encountered in the immediate vicinity of any given policy—and shows that the spread of that distribution is a hidden but practically important dimension of policy quality. Their work motivates a learning-stability metric based on per-window return variance that is used directly under the noise framing.

Seed counts and statistical reporting. The literature spans a wide range of seed counts: from only 3 runs in Li [23] to 5 in Zhu et al. [45], 10 in Nisioti et al. [27], 15 instances per setting in Girardin [10], 20–30 in Khatami [21], and 24 in Ecoffet et al. [6]. Agarwal et al. [1] argue for 20–50 independent runs as a minimum for reliable statistical conclusions and warn specifically against reporting point estimates alone—their recommendation of the interquartile mean (IQM) and paired-seed evaluation is increasingly adopted.

Statistical methods and interaction analysis. Across the corpus, three reporting conventions dominate: mean $\pm$ standard deviation (universal), pairwise non-parametric Mann–Whitney U tests [27, 6], Friedman–Nemenyi for multi-algorithm ranking with critical-difference diagrams [15, 21], and Cohen’s d alongside p -values ($d = 0.2$ small, $d = 0.5$ medium, $d = 0.8$ large) [28, 21]. Bayesian analysis—which quantifies evidence via Bayes factors and handles interaction effects naturally—is strikingly absent from the entire corpus. This absence directly motivates the planned Bayesian-ANOVA analysis for the interaction question, which remains an open methodological gap in the literature.

Reproducibility. Code availability is near-universal across the surveyed papers; explicit seed listings, Docker containers and pre-registration are not. The pre-registered systematic review of EvolRL by Lin et al. [25] on the Open Science Framework sets a transparency precedent for the field, but no empirical EvolRL study in this corpus has yet followed suit.

Methodology summary. None of these methodological sources should be read as independent evidence about noise or sparsity robustness; each is cited here only for the parameter, metric or statistical recommendation it contributes.

Table 1: Key methodological conventions adopted from the literature.

Category	Recommendation	Source(s)
PPO learning rate	3×10^{-4}	Schulman et al. (2017); Zhu et al. (2023)
PPO clipping ϵ	0.2	Schulman et al. (2017)
GAE λ	0.95	Schulman et al. (2016)
CMA-ES pop. size λ	$4 + \lfloor 3 \ln n \rfloor$	Hansen & Ostermeier (2001)
Primary metric	Final return (last 10%)	Schulman et al. (2017); Ecoffet et al. (2022)
Stability metric	Per-window return var.	Rahn et al. (2023)
Noise transfer	Robustness ratio φ_{12}	Li (2020)
Seed count	20–50 independent runs	Agarwal et al. (2021)
Pairwise test	Mann–Whitney U	Nisioti et al. (2025); Ecoffet et al. (2022)
Multi-algo ranking	Friedman–Nemenyi	Hernández & Santana (2025); Khatami (2026)
Effect sizes	Cohen’s d with p -values	Nomura et al. (2023); Khatami (2026)

5 Observation Noise

The noise evidence relevant to this study is fragmented along several axes: by the noise model studied (additive fitness noise, multiplicative noise, decision-parameter noise, subtractive/outlier noise, observation noise, partial observability); by the algorithm variant (vanilla (μ, λ) -CMA-ES, active (μ_w, λ) -aCMA-ES, multi-objective MO-CMA-ES, LRA-CMA-ES, NES/NESHT, PPO, SAC, TD3); and by the domain (synthetic BBOB functions, quantum-control benchmarks, robotics, MuJoCo, Atari). The recent survey by Lin et al. [25] catalogues this fragmentation as a general feature of EvolRL empirical practice. None of the sources reviewed here studies observation noise jointly with reward sparsity, and none that uses an RL observation-noise model has a controlled head-to-head comparison between PPO and CMA-ES. The evidence is grouped by the mechanism it illuminates.

5.1 The Foundational, Theoretical Case for CMA-ES Noise Tolerance

The positive case that CMA-ES should be noise-tolerant is theoretical, built from the design papers already reviewed. Population-based averaging is

explicitly motivated as a noise-reduction mechanism [12]; and cumulation smooths per-generation stochastic fluctuations into a stable evolution-path signal [13, 14]. What is essential to note is that none of the three foundational CMA-ES papers actually tests noise. The case rests on the design purpose of each mechanism, not on an empirical noise study of CMA-ES.

5.2 Bounded Robustness: Vanilla CMA-ES Has a Noise Threshold

The most direct formal evidence for a CMA-ES noise boundary is Nomura, Akimoto and Ono [28], who explicitly tackle “whether the CMA-ES with default population size can solve multimodal and noisy problems” by introducing a learning-rate (c_m) adaptation that holds a constant signal-to-noise ratio (SNR) in the update. Two findings are load-bearing. First, the vanilla CMA-ES with a fixed learning rate has a well-defined noise threshold beyond which its performance collapses on additive noisy functions—the reported cross-over lies near $\sigma_n^2 = 10^6$, where LRA-CMA-ES continues to make progress while fixed-learning-rate CMA-ES stagnates. The paper exists precisely because, in the authors’ words, “the CMA-ES with even a relatively small λ (e.g., $\lambda = \sqrt{n}$) solves multimodal problems, through appropriate setting of c_m ,” but discovering “such an appropriate c_m is difficult in practice, as prior knowledge is often limited and hyperparameter tuning requires extremely expensive numerical investigations.” Second, their proposed learning-rate adaptation maintains a constant SNR and recovers robustness with the default population size. The principle established here—vanilla CMA-ES has bounded robustness to fitness noise, and the boundary is identifiable—is the empirical anchor for the noise-side mechanism diagnostics in our design. It does not, however, study observation noise, and explicitly limits its noise experiments to additive fitness noise.

A complementary line targets a structurally different noise form: irrelevant-feature noise, where high-dimensional state vectors carry task-irrelevant components the algorithm must learn to ignore. Gao et al.’s NESHT [8, 9] combines Natural Evolution Strategies with a hard-thresholding step so that only features whose gradient contribution exceeds a threshold enter the update; under $\times 20$ noise

dimensions ($20\times$ more irrelevant than relevant features) the algorithm maintains high performance where vanilla NES collapses. Their analysis attributes the effect to a gradient-estimator variance that scales with dimension d —a first-principles reason that induced parameter sparsity improves noise tolerance. The result is orthogonal to the present study (irrelevant-feature noise is not i.i.d. Gaussian observation noise), but reinforces the general principle that different types of noise lead to different ES-side robustness profiles, and motivates careful sampling of the noise model rather than treating "noise" as a single stressor. A still older KL-bounded ES variant, MORE [17], makes the same point from a different angle: with a multiplicative noise model on a Noisy Sphere, MORE converges where standard CMA-ES, NES and PoWER diverge—robustness is again mechanism- and noise-model-specific, not unconditional.

5.3 Additive Versus Subtractive Noise: Why the Direction of Noise Matters

Girardin [10] benchmarks the non-elitist (μ_w, λ) -aCMA-ES—exactly the variant used in continuous-control policy search—on the BBOB testbed under additive and subtractive heavy-tailed (half-Cauchy) noise across dimensions 2–40, with $\lambda = 4 + 3 \log n$ (matching the default) and $\sigma_0 = 2$. The headline result is structural rather than merely quantitative:

"We find CMA-ES to be more sensitive to subtractive noise than to additive noise" [10], with the mechanism given as "bad outliers do not play a significant role in the selection process of (CMA-)ES. In contrast, good outliers are attributed large weights for recombination and the algorithm is perturbed by random directions. This slows down the algorithm."

This is the clearest available statement of why a rank-based, recombination-based algorithm is not symmetrically noise-robust, and it has direct implications for how to monitor CMA-ES during training. A noise model that systematically inflates apparent fitness (subtractive in a minimisation problem—an apparent "good outlier") is far more damaging than one that deflates it, because the inflated individuals are the ones the algorithm gives large recombination weight to. The paper further reports a resampling–evaluation tradeoff that has its own threshold: simple median-of-three

resampling is "beneficial for high noise levels" but "detrimental for low noise levels" [10], with the cross-over lying between noise levels 0.1 and 0.3 in 20D. The implication for mechanism diagnostics is that any rank-based metric must distinguish "bad outlier" from "good outlier" regimes, since the downstream failure modes are different.

5.4 The Decision-Parameter Noise Evidence: Shir, Roslund and Rabitz

The only direct noise-robustness evidence available for a CMA-ES variant under decision-parameter noise is the multi-objective quantum-control study of Shir, Roslund and Rabitz [37]. They study MO-CMA-ES under additive Gaussian noise applied directly to the decision parameters of a laser-pulse-shaping control field at levels $\sigma_x^2 \in \{0.01, 0.02, 0.05, 0.075, 0.10, 0.15\}$, propagating non-linearly to the objectives through the underlying quantum dynamics—a noise model they explicitly distinguish from the additive-fitness-noise model used elsewhere. Their central finding is a warning for empirical comparisons generally: perceived Pareto fronts look like excellent approximations to the true front, but a posteriori noise-free reevaluation of the same archived solutions reveals severe clustering—the algorithm converges to a minimal set of points whose noise-induced scatter happens to span the perceived front rather than genuinely diverse optima. The mechanism is tied to elitism: noise-disturbed fitness values are systematically overvalued under the elitist $(1 + \lambda)$ selection kernel used by MO-CMA-ES, so noise-inflated individuals survive and come to dominate the population.

Two structural differences limit direct transfer of this evidence to the present study. First, MO-CMA-ES uses an elitist $(1 + \lambda)$ kernel; the (μ, λ) -CMA-ES used in continuous-control policy search discards the parent each generation, so Shir et al.'s overvaluation mechanism is explicitly tied to elitism and is not guaranteed to reproduce under comma selection. Second, the noise is on decision parameters of a physical quantum-control system, not on the state observations of a sequential RL agent, and the domain is not RL at all. What does transfer—and is genuinely useful—is the general conclusion that CMA-ES's headline robustness can mislead at the level of perceived performance, and that diversity loss via clustering is a concrete, measurable fail-

ure mode distinct from a simple performance drop. That motivates population diversity as a hands-on secondary diagnostic rather than an assumed constant.

5.5 Adjacent CMA-ES Noise-Handling Work

Beyond the studies already reviewed, the CMA-ES community has produced a body of variant work that tackles noise directly through additional evaluations (re-averaging and adaptive re-evaluation [5, 39], surrogate-assisted re-estimation) or through problem decomposition (co-operative co-evolution with landscape-aware grouping [41]). We do not review these in depth here because, on inspection, none of them targets an RL observation-noise model and none combines noise with reward sparsity—the two axes this study is built to cross. Their existence is noted only as background: the CMA-ES community treats noise as a non-trivial design problem requiring either extra evaluation budget or explicit decomposition, rather than something the default algorithm absorbs for free, which reinforces the bounded-robustness framing that Nomura et al. [28] and Girardin [10] formalise.

5.6 The PPO-Side Noise Evidence: Noisy Neighborhoods

On the gradient-based side, the noise evidence that transfers to PPO is Rahn et al. [32]. They introduce the return landscape as the mapping from policy parameters θ to the episodic return $R(\theta)$, and demonstrate—for SAC, TD3 and PPO on standard MuJoCo environments—that “a single gradient step or perturbation to the policy parameters often causes important changes in the return, even in settings where both the policy and the dynamics are deterministic.” The neighbourhood of any given θ is therefore noisy in the precise sense that a population of perturbed policies produces a wide-spread distribution of returns rather than a tight cluster. They show that this “noisy neighborhood” phenomenon is a hidden dimension of policy quality: two policies with the same mean return can differ substantially in their post-update return-distribution spread, and the wider-spread policy is more failure-prone and less robust to small environmental perturbations.

The relevance to this study is direct. PPO updates parameters because of the gradient signal, and if that signal lives inside a noisy neighbourhood, the per-update behaviour is itself a noisy variable. Rahn et al.’s finding justifies per-update gradient variance, gradient SNR and post-update return variance as legitimate diagnostic quantities for tracing PPO’s behaviour under observation noise. The paper does not, however, study CMA-ES, observation noise as a controlled variable, or its interaction with reward sparsity.

5.7 Observation Noise in Adjacent Domains

Both Li [23] (already cited for its robustness-ratio metric) and a small body of adjacent robotics work treat observation noise as a first-class concern but stop short of crossing into the sparsity axis. Li runs Random Search, CEM and a Genetic Algorithm as derivative-free baselines on CartPole with i.i.d. Gaussian observation noise $\beta \in \{0, 0.25, 0.5, 1\}$, holding the noise injection in the input to the policy. The headline empirical finding is the qualitative “train noisy, test better” transfer effect: a policy trained under a higher noise level than it is tested on can show $\varphi > 1$, while the converse shows $\varphi < 1$; Li reports values ranging from $\varphi \approx 2.36$ to $\varphi \approx 0.42$ depending on the train–test gap. The RL-ACT approach to Kalman-filter covariance tuning [11] applies PPO directly to online estimation of measurement-noise covariances and reports a 14.73% accuracy improvement over fixed-covariance filtering, using identical PPO defaults ($\varepsilon = 0.2$, $K = 10$ epochs, batch 256). Partial-observability UAV landing addresses the same concern from within RL: the RPO-LSTM approach couples a PPO variant with recurrent memory to absorb observation noise and sensor-dropout flicker, outperforming standard PPO and a classical EKF controller under the hardest combined conditions [2]. These sources confirm that observation noise is a recognised concern in adjacent communities; none of them compares PPO against an evolution strategy, and none of them combines noise with reward sparsity.

5.8 Synthesis

Taken together, the noise evidence has a clear shape. The theoretical case for CMA-ES’s noise tolerance (population averaging, rank invariance, cumulation)

is well-motivated by the design papers but, with the partial exception of Nomura et al., was untested empirically under any noise model in an RL context until very recently. The closest direct evidence—Shir et al. on decision-parameter noise, Girardin on additive and subtractive fitness noise, Nomura et al. on additive fitness noise with the bounded-robustness threshold—covers different noise models and different domains; each shows bounded, not unconditional, robustness, and each separates good-outlier from bad-outlier regimes. The PPO-side evidence (Rahn et al., Li) confirms that gradient-based policy methods traverse noisy neighborhoods of the return landscape in which a small update can produce a wide spread of returns, motivating per-update gradient diagnostics as direct mediators. The combination of these strands supports a qualitative observation—PPO should degrade faster than CMA-ES under observation noise—without establishing where CMA-ES’s own threshold sits under RL observation noise specifically.

6 Reward Sparsity

Reward sparsity has by far the strongest direct evidence base in this review, anchored on Ecoffet et al.’s PLOS ONE paper [6]. A wider body of work spans general RL sparsity-handling methods and EA-side approaches that exploit, restructure or substitute sparse rewards.

6.1 The Direct PPO vs. CMA-ES Sparsity Comparison

Ecoffet et al. report the same controlled comparison of PPO against CMA-ES on an abstract partner-choice task (a continuous-investment Prisoner’s Dilemma variant), varying the probability $p \in \{0.1, 0.2, 0.5, 1.0\}$ of encountering a cooperative partner—a “significant event”—with episode length scaled as $T = 100/p$ to hold the expected number of meetings constant. The payoff follows $P(x_\bullet, x_i^+) = ax_\bullet + bx_i^+ - \frac{1}{2}x_\bullet^2$ with $a = b = 5$, giving a cooperative optimum at $x_\bullet = 10$ (payoff 50) and a defective equilibrium at $x_\bullet = 5$. Two PPO architectures are compared against CMA-ES: a small PPO-MLP (33 parameters) and a substantially over-parameterised PPO-DEEP (133,894 parameters), against CMA-ES’s 34-parameter genome, with 24 independent runs per condition and two-

tailed Mann–Whitney U testing. PPO is run with $\gamma = 1.0$ (no discounting, since episodes are not to be hurried), 10 epochs, batch 4000, minibatch 128; CMA-ES uses the default $\lambda = 14$ (i.e., $4 + \lfloor 3 \ln 34 \rfloor$) and $\sigma_0 = 1.0$.

Under dense reward ($p = 1.0$) the three methods are statistically comparable: PPO-MLP reaches median 45.58 (MAD 1.86), PPO-DEEP median 46.99 (MAD 13.04), CMA-ES median 47.64 (MAD 4.72) on re-evaluation across 1000 episodes, with most runs above 40. Under sparse reward ($p = 0.1$) the picture inverts sharply: PPO-MLP collapses to $\approx 24.9 \pm 4.2$, PPO-DEEP reaches only $\approx 35.7 \pm 5.2$ despite carrying roughly $4,000\times$ more parameters, while CMA-ES is essentially unaffected at $\approx 46.2 \pm 3.2$ [6], with the Mann–Whitney U test reporting $p < 0.0001$ at every sparsity level for CMA-ES against either PPO variant.

6.2 The Stated Mechanisms

Ecoffet et al. identify the PPO-side mechanism in language that our diagnostic framework is built to quantify directly: “when significant events are rare, gradient information is also scarce, making it difficult for policy gradient search methods to find an optimal policy, with or without a deep neural architecture” [6].

The CMA-ES-side mechanism is stated with equal directness and constitutes the strongest available articulation of why direct policy search should be invariant to event rarity:

“The robustness of the direct policy search method can be expected as the sequential and temporal aspects of the task is lost within one evaluation. As long as the evaluation time is long enough to sample the whole population of relevant partners, there is no cost nor change in the algorithm dynamics.” [6]

The paper is explicit that this differs fundamentally from sparse rewards generally:

“it may be tempting to relate the problem of rare significant events with that of sparse rewards... however, they differ fundamentally as significant events may be rare, but eventually occur. This is not the case with sparse rewards, which occurrences are conditioned by the policy itself.” [6]

6.3 The Fixed-Length Design Question: Why the Ecoffet Result Is Stronger Than It Looks

Ecoffet et al. scale episode length as $T = 100/p$, holding the expected number of significant events—and hence expected total reward—constant across sparsity conditions. This controls for one confound (overall reward budget) but introduces another: total reward magnitude is fixed across sparsity levels, so what is isolated is event rarity specifically, not the joint effect of rarity plus shrinking return magnitude. The strong CMA-ES invariance claim therefore rests on a design that structurally cannot produce a return-shrinkage regime. A fixed-episode-length design—adopted by the target study—is the more stringent test of CMA-ES’s mechanisms, since it allows expected return to fall as sparsity rises and selection pressure to weaken accordingly.

6.4 Synthesis and Threats to Validity

This is the one factor, of the two under study, with a controlled, statistically validated, mechanism-annotated head-to-head comparison directly between PPO and CMA-ES. Its two open edges for the target study are precisely a fixed-episode-length design and a continuous-control (MuJoCo) rather than abstract discrete-choice domain.

However, the single-anchor nature of this evidence warrants explicit acknowledgement. The Ecoffet result is the *only* controlled comparison of PPO against CMA-ES under sparse reward in the literature, and it uses PPO hyperparameters that deviate substantially from the MuJoCo defaults that the target study will employ. Specifically, $\gamma = 1.0$ (no discounting) maximises the difficulty of credit assignment for PPO, because rewards from distant timesteps are not exponentially decayed; this is a known vulnerability of policy-gradient methods with long horizons [34], and it may exaggerate PPO’s collapse under sparsity relative to what would be observed with a standard $\gamma = 0.99$ discount. The batch size of 4000 and minibatch of 128 also deviate from the MuJoCo-standard $T = 2048$ and minibatch 64 [35]; these choices may interact with the sparsity manipulation in ways that are not immediately transparent. Furthermore, the experiment has not been independently replicated, and the abstract continuous-investment domain, while internally controlled, has no direct

equivalent in the continuous-control benchmarks that dominate RL empirical practice.

The direction of these biases cuts against the review’s own framing. If Ecoffet et al.’s $\gamma = 1.0$ setting overstates PPO’s sparsity fragility, then the target study’s standard- γ , fixed- T design may plausibly find a weaker PPO degradation than predicted. This would not invalidate the factorial design—indeed, it would produce an informative result about the sensitivity of the PPO–CMA-ES gap to hyperparameter and domain choices. But it does mean that the directional expectation should be read as a hypothesis *from* the Ecoffet anchor, not as established fact, and that the target study is best understood as a strict re-test of the sparsity claim under more standard RL conditions.

The review mitigates this single-anchor fragility in two ways. First, it never presents Ecoffet as established truth; it presents it as the hypothesis-generating experiment, which is legitimate proposal logic. Second, Section 7 supplies convergent indirect evidence for the gradient-starvation mechanism from independent groups: Tse’s “zero gradients everywhere” formulation [38], CMA-EPPS beating DDPG on terminal-only reward [44], and CMA-MAPPO’s existence as a hybrid motivated precisely by PPO’s sparsity weakness [21] all point in the same direction.

6.5 The Wider Sparsity Literature

A substantial set of additional sources treats sparsity from one side or the other. The shared intuition on both sides is that gradient signal scarcity is the bottleneck; the mitigation strategies either reduce the sparsity (shaping, demonstrations, behaviour-space preprocessing) or sidestep it entirely (preference-based or direct policy search).

RL-side mitigations

Inner-reward shaping in sparse-reward games [43], behaviour-space discovery as a way of finding “regions where reward is learnable” [30], reward learning from very few demonstrations [7], and advantage–value difference reward shaping adapted specifically to a PPO perspective [24]. All four strategies share the goal of densifying the learning signal so that gradient information is available more often than the environmental reward itself supplies it.

EA-side alternatives

Bilevel evolutionary reward shaping [40]; sparse-cosine-optimised policy evolution on Atari [29]; and the preference-based CMA-EPPS framework for robotics confrontation, where the reward derives only at the end of an episode—with the authors noting that “conventional deep reinforcement learning such as Deep Deterministic Policy Gradient performs poorly for sparse reward problems” [44] and that CMA-based preference search performs better in learning speed, winning rate and robustness than reward-based methods despite a smaller policy network. Tse’s Bayesian sparse-sampling thesis [38] uses CMA-ES as a black-box policy optimiser precisely because the model-based RL loss surface “has zero gradients everywhere”—a first-principles reason that direct search can dominate gradient ascent in sparsity-like regimes. These strategies share the goal of avoiding gradient estimation entirely, so that sparsity in reward no longer translates into sparsity of learning signal.

What does not appear in the literature

A controlled, statistically instrumented, mechanism-annotated comparison of PPO against CMA-ES on continuous control with a fixed episode length and a noise axis crossed in. The wider literature provides the surrounding intuition that gradient starvation is the dominant PPO-side failure mode and that population-based methods sidestep it by virtue of treating episodes as atomic. The set of mitigations surveyed above—shaping, demonstrations, behaviour-space preprocessing on the RL side, preference-based search and bilevel reward shaping on the EA side—further demonstrates that the sparsity problem is felt widely enough to motivate entire subfields on both sides of the gradient divide, yet the two sides’ approaches have not been brought into direct, factorial, mechanism-instrumented contact on the same continuous-control benchmark.

7 Hybrid Methods and the ERL Design Space

A rapidly growing family of hybrid methods sits between PPO and CMA-ES, attempting to inherit strengths of both paradigms. Although none of

them directly tests both observation noise and reward sparsity together, their existence and design rationale are useful context for this study.

CMA as a step-size controller for PPO. The PPO-CMA combination uses CMA-ES to manage the step-size (trust region) of PPO updates, leveraging CMA-ES’s adaptation to keep PPO’s policy updates within a productive region [18]. A related line accelerates an evolution strategy by inserting PPO’s clipping mechanism into the ES update loop [42], and a third combines PPO with a broader evolutionary strategy in a single hybrid loop, often as an off-policy correction to the on-policy gradient estimate [36].

Interleaved ES and gradient-based search. The CEM-RL family interleaves a cross-entropy method with DDPG gradient steps [31]; the hard-thresholding-meets-ES approach of Gao et al. [8, 9] uses a hard threshold on candidate fitnesses inside an ES update to accelerate convergence on RL benchmarks; and bounded-support evolution strategies for policy refinement exploit explicitly bounded perturbations to guarantee safety constraints while still allowing ES-style exploration [16]. OpenAI-ES [33] established the broader precedent that evolution strategies can compete with RL on continuous-control benchmarks, albeit under dense rewards and without a combined noise-sparsity treatment.

ES for exploration under sparse reward. The most directly relevant hybrid is Khatami’s CMA-MAPPO [21], which integrates CMA-ES with multi-agent PPO specifically to enhance exploration in sparse-reward environments—a clear statement, from inside the hybrid literature, that the sparsity side of the PPO/CMA-ES contrast is the recognised weak point of pure PPO. CMA-EPPS applies a preference-based CMA-ES variant to terminal-only sparse-reward robotics confrontation and outperforms DDPG baselines [44].

The survey-level view. Three recent surveys situate the broader design space. Bai, Cheng and Jin [3] and Zhu et al. [45] cover ERL as a field, identifying common algorithmic patterns (population-based training, gradient-EA hybrids, neuroevolution) and reporting the synchronous benchmarking conventions used by ERL evaluations. Li et al. [22] organise the literature into three branches—EA-assisted optimisation of RL, RL-assisted optimisation of EA, and synergistic optimisation—and

note explicitly that EAs “show good robustness and convergence properties, displaying resistance to noise and uncertainty” [22], while RL provides the sample efficiency that pure EAs lack. The recent systematic review by Lin et al. [25], which the authors pre-registered on the Open Science Framework, doubles as a methodological precedent for the present study and explicitly catalogues the field’s reproducibility gaps (no pre-registered empirical EvoRL study, sparse Docker/container coverage, inconsistent seed listing). The Deep RL vs. ES comparative survey [26] frames the contrast across dimensions (parallelism, exploration, dynamic environments, multi-agent learning), explicitly cataloguing “dealing with sparse rewards” as an open challenge for both families and partial observability as a setting where both RL and ES require architectural support. All surveys confirm that noisy and sparse-reward robustness are recognised but unsolved problems for the combined regime.

8 The Interaction Gap

Every primary source reviewed above does not test observation noise and reward sparsity jointly. Table 2 categorises this .

The two papers that most directly anchor this study’s opposing halves—Ecoffet et al. for the sparsity-side PPO/CMA-ES comparison and the noise cluster (Girardin, Nomura, Shir) for the CMA-ES noise evidence—do not overlap in any dimension beyond the algorithm lineage and the general mechanism vocabulary (rank-based selection, population averaging, cumulation). The sparsity side tests one gradient-based method and one gradient-free method in a non-continuous-control domain under a budget that scales with sparsity; the noise side tests variants of CMA-ES on synthetic fitness noise models or on non-RL domains with an elitist algorithm variant. The two halves use neither the same benchmarks nor the same noise abstraction, and no source in this review crosses both axes.

This is not an incidental gap. It reflects a genuine split in the underlying literatures—ES-focused noise theory built on non-RL optimisation benchmarks; RL-focused sparsity comparisons built on episodic reward structures—and the boundary between them is what the target factorial design is positioned to bridge, on the same benchmark family for both factors and both algorithms.

9 Conclusion

The two papers that anchor the study’s opposing halves—the sparsity-side PPO/CMA-ES comparison and the noise cluster—do not share a single dimension beyond algorithm ancestry and general mechanism description (rank-based selection, population averaging, cumulation). The sparsity group evaluates one gradient-based and one gradient-free algorithm in a non-continuous-control domain with a reward budget defined by the sparsity; the noise group evaluates variations of CMA-ES either in synthetic fitness noise domains or non-RL environments with a different elitism-based algorithm variation. The two halves do not share either benchmark or noise abstraction, and none of the sources reviewed address both factors in any capacity.

This is unsurprising in context: the literature on ES methods’ noise properties is focused on non-RL domains, while RL-focused literature on reward sparsity only compares algorithms with similar mechanistic roots (PPO and CMA-ES). These areas were bridged by the target study’s factorial design, which examines the interaction of the two solution qualities using the same sets of algorithmic variations (to the extent of the primary literature’s availability for the two algorithms).

The evidence base for the current study was split cleanly between the two factors in question. Reward sparsity has a direct, controlled comparison between PPO and CMA-ES with annotated mechanisms ([6]), whose limitations (reward-constant episode length, abstract continuous-investment domain as opposed to continuous-control robotic domains) clearly define the target study’s scope. In turn, the observation noise factor has no direct comparison to which CMA-ES’s noise properties ([37]) could be linked, and the closest evidence for PPO’s abilities ([32, 23]) describes the properties of the domain in which the algorithm was tested (noisy neighborhood of the return landscape, transfer-of-robustness ability) without a direct comparison to an evolution strategy. The design of experiments that the current study conducts is positioned to fill this evidence gap.

The theoretical basis for CMA-ES’s noise properties ([13, 14, 12, 4]) is similarly positioned to complement the findings of the primary sources by targeting the same set of noise properties (independent variations of additive, subtractive, and

Table 2: Coverage of noise, sparsity, and their combination across the primary sources.

Source	Noise	Sparse.	Comb.	PPO vs. CMA-ES
Hansen & Ostermeier '96 [13]	No	No	No	– (ES only)
Hansen & Ostermeier '01 [14]	No	No	No	– (ES only)
Hansen et al. '03 [12]	No	No	No	– (ES only)
Beyer & Sendhoff '17 [4]	No	No	No	– (ES only)
Nomura et al. '23 [28]	Yes	No	No	– (CMA-ES only)
Girardin '25 [10]	Yes	No	No	– (CMA-ES only)
Shir et al. '09 [37]	Yes	No	No	– (MO-CMA-ES)
Rahn et al. '23 [32]	Yes	No	No	– (PPO-side)
Li '20 [23]	Yes	No	No	– (derivative-free)
Schulman et al. '17 [35]	No	No	No	– (PPO only)
Schulman et al. '16 [34]	No	partial	No	– (PPO stack)
Ecoffet et al. '22 [6]	No	Yes	No	Yes
ERL surveys [3, 45, 26, 22]	partial	partial	No	discussed, not tested
This study (proposed)	Yes	Yes	Yes	Yes

decision-parameter noise) using the same algorithm variations (while using the CMA-ES-specific variations of noise budget). Each of the five papers that contributed to the current study’s design explicitly states that it only evaluates solutions’ properties in isolation. This evidence gap—properties of solutions in isolation, for either algorithm, and of the two algorithms compared to each other—is what the current study’s factorial design is positioned to address, with the supporting findings on the mechanics of sparsity and observation noise described in the review’s corresponding sections providing the necessary contextualization. The language of causality vs. description used in the discussion of CMA-ES vs. PPO, respectively, is also a methodological complement, produced by the noise cluster’s anchoring contribution ([37]) which explicitly evaluates the mechanisms of noise resistance at the level of solution quality.

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